

Practical Work 7: Sudoku on Google App Engine

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1 Introduction

Google App Engine (GAE) is a Platform as a Service (PaaS) offering that allows developers to build and run applications on Google's infrastructure. This practical work demonstrates how to build a simple Sudoku web application using Python (Flask) and prepare it for deployment on GAE.

2 Project Structure

The project follows the standard structure required by the GAE Python runtime environment:

- `app.yaml`: The configuration file that defines the runtime environment and handlers.
- `main.py`: The core application logic utilizing the Flask micro-framework.
- `requirements.txt`: Lists dependencies (Flask, Gunicorn) to be installed by GAE.
- `templates/index.html`: The frontend HTML template for rendering the Sudoku board.

3 Implementation Details

3.1 Configuration (`app.yaml`)

This snippet defines the Python 3.9 runtime and the entry point for the application using Gunicorn.

```

1 runtime: python39
2 entrypoint: gunicorn -b :$PORT main:app
3
4 handlers:
5   - url: /*
6     script: auto

```

3.2 Application Logic (main.py)

We use a list of lists to represent the Sudoku board. The `index` function handles both GET requests (displaying the board) and POST requests (validating the user's solution).

```

1 from flask import Flask, render_template, request
2 app = Flask(__name__)
3
4 # Simplified Board Logic
5 @app.route('/', methods=['GET', 'POST'])
6 def index():
7     board = PUZZLE_BOARD
8     message = ""
9     if request.method == 'POST':
10         # Logic to validate user input against SOLVED_BOARD
11         # Sets message to "Congratulations" or "Incorrect"
12         pass
13     return render_template('index.html', board=board, message=
        message)

```

3.3 Frontend Interface (templates/index.html)

The HTML template utilizes Jinja2 templating engine to render the 9x9 grid dynamically. It uses nested loops to create input fields for each cell and submits the user's data via a POST form.

```

1 <!DOCTYPE html>
2 <html>
3 <body>
4   <h1>Sudoku on Google App Engine</h1>
5   <form method="POST">
6     <table>
7       {% for r in range(9) %}
8       <tr>
9         {% for c in range(9) %}
10        <td>
11          <input type="text" name="cell-{{r}}-{{c}}"

```

```

12         value="{{_board[r][c]_if_board[r][c]_!=
    _0_else_'_'_}}"
13         maxlength="1">
14         </td>
15         {% endfor %}
16     </tr>
17     {% endfor %}
18 </table>
19 <button type="submit" class="btn">Check Solution</button>
20 </form>
21 <div class="msg">{{ message }}</div>
22 </body>
23 </html>

```

4 Setup and Execution

Since modern Linux distributions adopt PEP 668, we cannot install Python packages system-wide. Therefore, we set up a virtual environment to run the application locally.

4.1 Environment Setup

We executed the following commands to prepare the environment and install dependencies:

```

1 # 1. Create a virtual environment
2 sudo apt install python3-venv -y
3 python3 -m venv venv
4
5 # 2. Activate the environment
6 source venv/bin/activate
7
8 # 3. Install required libraries
9 pip install flask gunicorn

```

4.2 Running the Application

After the setup, we started the Flask development server:

```

1 # Run the application
2 python3 main.py

```

The terminal output confirmed that the server was running on <http://127.0.0.1:8080>.

4.3 Results

We verified the gameplay by accessing the local web server. Figure 1 displays the Sudoku game interface where users can input numbers and check their solutions.

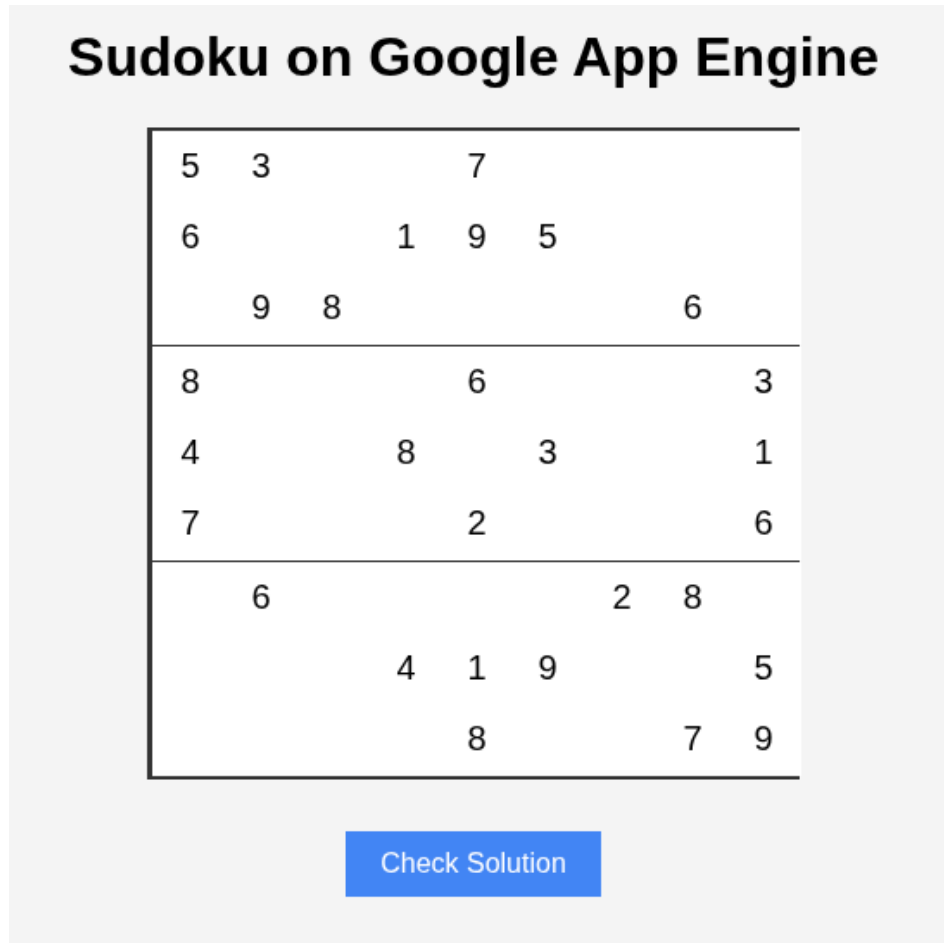


Figure 1: Sudoku Game Interface running on Local Development Server

5 Conclusion

In this practical work, we successfully structured a Python web application for the Google App Engine platform. By using Flask, we separated the logic (Python) from the presentation (HTML templates).

- The `app.yaml` file is crucial for GAE to understand how to serve the application.

- The PaaS model allows us to focus on the code (`main.py`) without worrying about the underlying infrastructure setup.